Paul Parisot

Software Engineer - Core C++ & Systems Programming

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About

Systems-focused software engineer with strong foundations in C/C++, Go, and Linux environments.
Experienced in building custom interpreters, reliable data parsers, and low-level modular architectures.
Rigorous developer with a focus on code quality, memory safety, and structural precision over raw speed.

Skills

Core Languages: C, C++ (Modern standards C++20/23), Golang, Haskell, OCaml, Python
Environment & Formats: Linux (Systems Programming), XML (Parsing & Validation), Git toolchain
Infrastructure: DevOps (Docker, Kubernetes), CI/CD pipelines, Unit Testing frameworks

Professional Experience

Gno.land - Junior Software Engineer (Open-Source System)

- Developed 6+ core packages in Gnolang (Go variant), including architectural components and compliance modules.
- Diagnosed and resolved **6 critical runtime crash bugs**, contributing over **10k lines of robust code**.
- Wrote extensive integration test suites under strict, zero-tolerance CI/CD and peer code review policies.

Faurecia / Forvia - Software Engineer (Embedded Demo)

- Developed **C++ network modules** for embedded communication, ensuring reliable low-level data exchange.
- Engineered a real-time navigation and GPS simulation application showcased globally at **CES 2024**.
- Designed a high-reliability GPS path generation engine focusing on exception handling and algorithmic precision.
- Delivered production-ready components, praised by management for architectural clean code and robustness.

Epitech / IONIS - Technical Assistant & Computer Science Instructor

- Taught system-level programming (**C, C++, Assembly, Haskell**) to over **400 students**; enforced strict coding standards.
- Documented technical guidelines, created clear educational content, and structured advanced assessment material.
- Collaborated closely with faculty to track student performance, ensuring high quality and rigorous methodology.

Technical Projects

Multi-format Data Parser – XML, JSON & Markdown

Built a strictly-typed abstract syntax tree (AST) parser and cross-encoder from scratch using functional paradigms.
Implemented data validation, structure compliance checkouts, and rigorous serialization for the **XML format specification**.

Skills: Haskell, OCaml, XML Specifications, Parsing Theory, Data Validation, Functional Types

rodox – Modular Systems Engine under Linux

Designed a modular systems architecture in C/C++ running natively in a **Linux environment**.
Leveraged Modern C++26 metaprogramming concepts to optimize compile-time introspection, memory layouts, and plugin linking.

Skills: C++, Linux OS, ECS Architecture, Metaprogramming, Custom Memory Management, Runtime Dynamic Linking

ZZZcript – Statically-typed Functional Language

Developed a full lexer, parser, and recursive AST evaluation engine from scratch.
Focused on strict error reporting and predictable evaluation semantics to avoid runtime memory overhead.

Skills: Go, Language Design, Interpreter Architecture, Type Checking, Deterministic Systems

Education

Epitech, Paris 2022 – 2027
Master's Degree in Information Technology (Level-7 RNCP Certification)
Specialization: Expert in Software Engineering

McGill University, Montréal 2025 – 2026
Certificate in Management
Core coursework: Business Administration, Organizational Management, Functional Accounting